

Integrated Optimization of Urban Morphology and Energy Performance through Deep Reinforcement Learning: A Block-Scale Application

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Abstract—Integrated optimization of urban morphology and energy performance is critical for decarbonization but is hindered by the separate and inefficient nature of traditional optimization methods. This paper proposes a novel Deep Reinforcement Learning framework to co-optimize UM and EP at the block scale. A Deep Neural Network (DNN) surrogate model, trained on a parametrically generated dataset, rapidly predicts Energy Use Intensity (EUI), Energy Generation (EG), and Sunlight Hours (H) from 12 morphological factors. A Deep Deterministic Policy Gradient agent then interacts with this surrogate to learn optimal design policies. Three specialist agents were trained with different reward weights (Balanced, Saving, Generation) and benchmarked against NSGA-II. The DRL framework substantially outperformed NSGA-II in solution quality (Hypervolume and IGD). Crucially, the Energy Saving agent did not merely identify a trade-off; it discovered a dominant solution region. This non-obvious strategy, defined by low building density and high open-space/sky-view factors, simultaneously minimized *EUI* while maximizing both *EG* and *H*. The DRL-DNN framework transforms optimization into an intelligent, goal-directed learning process, capable of uncovering superior, interpretable, and dominant design strategies missed by traditional algorithms.

Index Terms—Multi-Objective Optimization, Urban Morphology, Energy Performance, DRL.

I. INTRODUCTION

In the context of global strategies for carbon peaking and carbon neutrality, deep decarbonization of urban energy systems emerges as a central issue for sustainable development [1, 2]. Cities, as primary energy consumers, exhibit a strong and intricate relationship between urban morphology (UM) and energy performance (EP) [3-5]. However, current planning practices often treat urban design and energy system planning as separate fields, leading to significant energy inefficiencies and wasted renewable energy potential [6, 7]. Thus, developing a scientific framework for the integrated optimization of urban morphology and building energy performance has become a critical frontier challenge at the intersection of architecture, energy studies, and urban planning [8].

A. Motivations

To address this challenge, computational design methods based on parametric modeling and multi-objective optimization (MOO) algorithms have become mainstream [9, 10]. Evolutionary algorithms, exemplified by NSGA-II, are widely used to explore the design space. Their goal is to identify the optimal Pareto front among various conflicting objectives, such as energy consumption[11], energy generation potential[12], and daylighting[13]. However, this prevailing paradigm assumes a core premise: a trade-off necessarily exists between these optimization objectives. The core motivation of this study challenges the default assumption that trade-offs are the only inevitable outcome in the high-dimensional, nonlinear, and strongly coupled complex design space formed by urban morphology. Is it possible that non-obvious regions of optimal solutions exist, which traditional black box search algorithms overlook? In these specific regions, key performance indicators might achieve synergistic win-win outcomes. Current optimization methodologies clearly lack an intelligent decision-making mechanism capable of autonomously learning and proactively identifying such non-explicit strategies. This study, therefore, aims to explore a new methodological paradigm that not only searches the solution space but also learns strategies, thereby uncovering new design knowledge that transcends the traditional trade-off paradigm.

B. Related Work

Currently, the mainstream approach for addressing multi-objective optimization (MOO) problems in urban design relies on search-based algorithms [14, 15], particularly evolutionary algorithms like NSGA-II [16]. These algorithms primarily explore

complex problem spaces to identify trade-off solutions among conflicting performance metrics, such as energy efficiency and cost [17]. Despite its widespread application, this search-centric research approach faces three fundamental limitations. First, the computational cost is high. Evolutionary algorithms require iterative performance evaluations of numerous individuals or design schemes within a population. These evaluations, such as energy consumption simulations based on EnergyPlus, are inherently time-consuming [18, 19]. As previous studies highlight, when the dimensionality of the design space increases—leading to the curse of dimensionality—the time required to find a convergent Pareto front can extend from several days to weeks [20]. Second, evolutionary algorithms inherently function as black-box models [21]. Their random search mechanisms, such as crossover and mutation, do not fully exploit the underlying physical logic of the problem. Consequently, designers are presented with a cloud of Pareto-optimal solutions, making it challenging to derive interpretable and transferable design strategies [22, 23]. Third, the mechanisms of NSGA-II aim to maximize solution diversity and clearly outline the solution space's boundaries [24], specifically the Pareto front. However, these mechanisms inadvertently penalize solutions clustered in the same region, hindering the algorithm's convergence and its ability to effectively explore potentially dominant solution regions hidden within the space [25].

To overcome the aforementioned limitations, this study introduces Deep Reinforcement Learning (DRL), a fundamentally different learning paradigm [26]. DRL combines the nonlinear feature representation capabilities of deep learning with the sequential decision-making framework of reinforcement learning. This approach transforms static optimization problems into a strategy-learning process. Instead of conducting blind searches, the agent learns optimal strategies through interaction with the environment, utilizing a trial-and-error method. Deep reinforcement learning has achieved considerable success in complex, high-dimensional decision tasks such as robotic control (e.g., precise robotic arm grasping) [27], strategy games (surpassing human performance in complex games) [28], natural language processing [29], and autonomous driving (navigating continuous dynamic environments) [30]. These tasks share a core challenge: making sequential decisions in high-dimensional, continuous state and action spaces. This challenge mirrors the one faced in urban morphology optimization, where designers must make optimal decisions involving 12 continuously variable morphological factors. However, when applying deep reinforcement learning to optimize urban morphology, which involves high-dimensional and continuous action spaces, selecting the appropriate algorithm is essential. Value function-based methods, such as DQN [31], output Q-values for discrete actions and thus cannot directly handle continuous morphological parameters like building orientation and height. Therefore, this study selects the Deep Deterministic Policy Gradient (DDPG) algorithm. DDPG [32], as an advanced method based on the Actor-Critic framework, has been proven to handle continuous control problems efficiently and stably. Its Actor network learns a deterministic strategy, while the Critic network evaluates the long-term value of the actions selected by the Actor. This structure makes it an appropriate technical approach to addressing the complex design optimization challenges in this study.

Although the learning paradigm of DRL shows considerable potential, its application in the interdisciplinary field of urban morphology-energy performance optimization remains unexplored. While DRL is increasingly used in building energy control [33, 34] and energy system operation [35, 36], its use as a design/optimization tool for urban morphology is nascent. Furthermore, its advantages as a paradigm—specifically its ability to discover dominant solutions—have yet to be demonstrated and validated compared with traditional search paradigms like NSGA-II.

C. Contributions and Paper Organization

To address the motivations mentioned above and fill the research gap, this study develops and validates a DRL-DNN integrated framework. We train three specialist agents, each with distinct preferences (reward functions), and rigorously compare their performance with the NSGA-II benchmark.

- 1) This study introduces a novel DRL-DNN ensemble framework for optimizing urban morphology, transforming it into a DRL policy learning challenge. The effectiveness of the DDPG algorithm in addressing complex design problems is rigorously demonstrated.
- 2) This study demonstrates the superior performance of the DRL framework over NSGA-II in solution quality, as evidenced by higher scores in HV and IGD metrics. DRL consistently shows better convergence and solution distribution compared to NSGA-II.
- 3) This study reveals a key discovery challenging the conventional trade-off concept: DRL agents discover a solution area missed by NSGA-II. This area emphasizes strategies like low building density and high sky visibility, optimizing EUI, EG, and H objectives synergistically.
- 4) This study presents an interpretable design strategy that outperforms black box optimization methods. Unlike NSGA-II, the DRL agent generates clear and distinct design patterns, revealing the nonlinear link between design factors and performance.

The subsequent sections of this paper are organized as follows: Section II elaborates on the construction of the proposed DRL-DNN integrated framework, detailing the parameterized modeling, performance simulation, DNN surrogate model, and the implementation of the DDPG algorithm. Section III presents the background of the case study and provides an in-depth comparison and analysis of optimization results across multiple scenarios. Section IV discusses the core findings, methodological significance, and limitations of this study. Finally, Section V provides a summary of the entire paper.

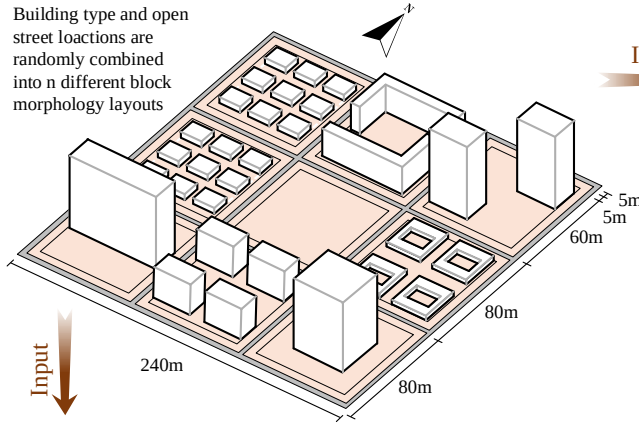
II. METHODOLOGY

This paper proposes a DRL-based framework for the automated and intelligent design of energy-efficient urban blocks. As illustrated in Fig. 1, the framework first uses a parametric model to generate a performance dataset linking UMFs to key EPIs.

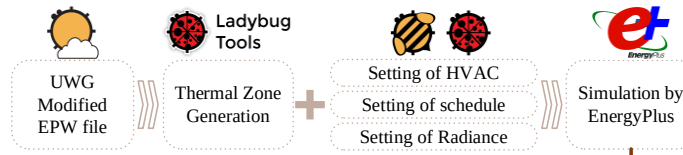
A DNN is then trained on this dataset to act as a computationally efficient surrogate model for rapid performance prediction. Finally, a DDPG agent interacts with this surrogate model to learn and generate optimal design strategies that reduce negative environmental impacts, yielding substantial benefits consistent with sustainable development goals.

I Model Generation

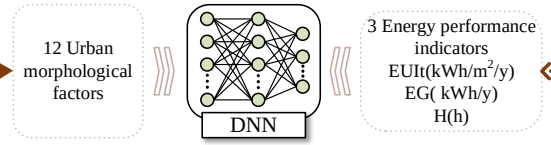
Building type and open street locations are randomly combined into n different block morphology layouts



II Performance Simulation



III DNN-based Surrogate Model



IV DDPG Framework

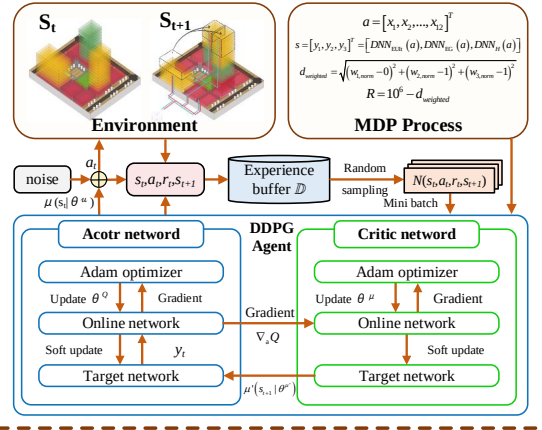


Fig. 1. Overview of proposed method

A. Case Study and Parametric Modeling

The selected research area is Jianhu County, located in Jiangsu Province in the eastern coastal region of China. The county is situated between 33°16' and 33°41' N latitude and 119°33' and 120°05' E longitude, with an average elevation of 2 m. It has an administrative area of approximately 1160 km² and a population of around 600,000, resulting in a population density of 520 persons/km². According to the Chinese Code for Thermal Design of Civil Buildings, Jianhu lies within the hot-summer and cold-winter climate zone and is classified as having a humid subtropical climate under the Köppen-Geiger system. The climate necessitates both building heating and cooling, as the average dry-bulb temperature is 27.1°C in the hottest month (July) and 1.6°C in the coldest month (January), with an annual average of 15.0°C. Furthermore, the region presents favorable conditions for solar energy utilization, receiving an annual solar radiation of 1250 kWh/m² with approximately 1800 hours of sunshine per year.

To generate a diverse dataset linking urban form to energy performance, a parametric modeling approach was implemented on the Grasshopper platform. The core of this approach involves defining a standard urban block subdivided into a 3x3 grid, creating nine land units. Based on an analysis of the existing urban fabric, a set of representative residential building prototypes (e.g., point, slab, and courtyard types) were defined. In the parametric model, one land unit is designated as open space, while the remaining eight units are populated with these building prototypes to generate a wide variety of block morphologies. This parametric generation process is controlled by twelve selected UMFs. These include key geometric indicators such as Floor Area Ratio (*FAR*), Building Density (*BD*), Open Space Ratio (*OSR*), Average Floors (*AF*), Setback Diversity (*SD*), Shape Coefficient (*SC*), Average Perimeter-to-Area Ratio (*PAR*), Aspect Ratio (*AR*), and average Sky View Factor (*SVF*). Once the geometric model of the block is configured on the Grasshopper platform, these UMFs can be calculated using its basic tools. The calculation formulas for these morphological factors are presented in Table I. In addition to the factors in Table I, an Open Space Location Index (*OSLI*) is established to optimize the local wind, thermal, and light environments¹⁷. The *OSLI* is an integer variable (ranging from 0 to 8) representing the topological position of the open space within the 3x3 grid¹⁸. Furthermore, the Block Orientation Angle (θ), ranging from $[-45^\circ, 45^\circ]$, is included to regulate solar radiation.

The specific geometric parameters of the block layout (e.g., overall dimensions, setback rules) and the detailed definitions

TABLE I
CALCULATION FORMULAS OF THE MORPHOLOGICAL FACTORS

FAR	SD	AF	AR	SVF	BD	OSR	SC	PAR
$\frac{\sum_{i=1}^n S_i}{A_s}$	$h_{\max} - h_a$	$\frac{\sum_{i=1}^n S_i}{\sum_{i=1}^n f_i}$	$\frac{\sum_{i=1}^n H_i}{\sum_{i=1}^n W_i}$	$\int_{\theta=0}^{2\pi} \cos^2(\beta(R, \theta)) d\theta$	$\frac{\sum_{i=1}^n f_i}{A_s}$	$\frac{As - \sum_{i=1}^n f_i}{\sum_{i=1}^n S_i}$	$\frac{\sum_{i=1}^n c_i}{\sum_{i=1}^n h_i f_i}$	$\frac{\sum_{i=1}^n p_i}{\sum_{i=1}^n f_i}$

Note: n is the number of buildings; OR_i is the orientation of building i ; S_i is the total floor area of building i ; A_s is the area of block; f_i is the floor area of building i ; $h_{\max}$, h_a are the maximum and average building height of the block; H_i , W_i is the height and width difference of the Canyon. c_i is the surface area of building i ; p_i is the perimeter of building i ; β is the angle from the center point to the maximum obstacle height at a maximum distance equal to the constant search radius (R).

of the building prototypes used to generate the dataset are comprehensively documented in the Supplementary Material(Section S-1).

B. Performance Simulation and Optimization Objectives

This study assesses the building energy consumption and solar potential of residential blocks using three key Energy Performance Indicators (EPis): total Energy Use Intensity (*EUI*), Energy Generation (*EG*), and Sunlight Hours (*H*). These indicators serve as the foundation for developing strategies that mitigate environmental impacts and enhance energy efficiency in urban planning. *H*, which indicates the duration of solar radiation exposure, was simulated using the Sunlight Hours Analysis module in Ladybug for a typical coldest day (January 20th, 8:00–16:00). Continuous test points were set at 1 m intervals along the south-facing ground-floor windowsills (0.9 m above ground), and the Radiance engine calculated solar trajectories and shading; the average sunlight hours across all test points represent the block's overall daylighting performance. *EUI*, a core indicator of energy-saving performance, was simulated by constructing a Honeybee energy model based on the EnergyPlus engine, with parameters set according to Chinese standards JGJ 134-2010 and GB 50736-2012 (details in Table II). This simulation applied a stratified thermal zoning strategy, set the building function uniformly to residential, and used the Urban Weather Generator (UWG) to convert standard weather data into urban microclimate data to account for the UHI effect. *EG*, representing the block's potential for renewable energy development, was calculated based on annual rooftop solar radiation from a coupled Ladybug-Radiance simulation. This model assumes a polycrystalline silicon PV system (17% efficiency, 85% DC/AC conversion) under standardized conditions (90% roof coverage, 0° tilt) to ensure a comparable analysis. PV potential was then assessed against a threshold of 800 kWh/m²/year. Finally, a Dragonfly-Ladybug toolchain was constructed to automate geometric information extraction and parameter passing, ensuring spatial consistency across all simulation models.

TABLE II
PARAMETERS AND SETTINGS FOR THE BUILDING ENERGY SIMULATION IN ENERGYPLUS

Classification	Parameter	Specification
System Settings	Weather Data	EPW files modified based on UWG urban microclimate simulations
	Simulation Period	Annual simulation (January 1 to December 31)
Envelope Thermal Performance	Wall U-value	0.8 W/(m ² ·K)
	Roof U-value	0.5 W/(m ² ·K)
	Floor U-value	1.5 W/(m ² ·K)
	Window Properties	U-value = 2.70 W/(m ² ·K) Solar Heat Gain Coefficient (SHGC) = 0.78
Window-to-Wall Ratio	North & South Facades	0.4
	East & West Facades	0.1
Internal Loads	Occupant Density	0.03 persons/m ² , Metabolic rate = 100 W/person
	Lighting Power Density	5 W/m ²
	Equipment Power Density	1.9 W/m ²
HVAC System	Ventilation Rate	30 m ³ /(h·person)
	Thermostat Setpoints	Heating: Dec 1 – Feb 28, 18°C; Cooling: Jun 15 – Aug 31, 26°C
	Infiltration Rate	1 ACH (Natural air leakage)

C. DNN-based Surrogate Model

In this study, a DNN is constructed as a predictive surrogate model, trained on a comprehensive simulated dataset to accurately predict the three target building performance indicators. DNN is a type of multi-layer perceptron capable of modeling complex, non-linear relationships by learning hierarchical feature representations from data. This study employs a multi-layer feedforward DNN architecture, where information flows unidirectionally from the input layer, through the hidden layers, to the output layer. In this architecture, input data is passed and transformed layer-by-layer through fully connected neurons, enabling the model to represent non-linear relationships within the data and ultimately produce a prediction.

The DNN training process consists of two key stages: forward propagation and backpropagation. In the forward propagation stage, the network processes input data to generate a prediction. During backpropagation, the prediction error is propagated backward from the output layer to the input layer. This process involves calculating the gradient of the loss function with respect to the network's weights, which allows for the iterative optimization of the model's parameters. The core of backpropagation is the efficient computation of this gradient, as shown in Equation (1) which provides a quantitative basis for precisely adjusting the DNN's weights. By optimizing the weights through backpropagation, the model's prediction error is effectively reduced, thereby improving its reliability and its ability to generalize to unseen data.

$$w = w + \eta \times \nabla w = w + \eta \times \text{backpropagation}(f(x)) \quad (1)$$

where η is the learning rate, and $f(x)$ is the objective function to minimize.

To ensure an accurate mapping between the network's inputs and outputs, hyperparameter tuning is a critical task. This study employs Bayesian Optimization (BO) to systematically search for the optimal hyperparameter configuration. The BO process

was configured to minimize the Mean Absolute Error (MAE) on a validation subset of the training data. To ensure the tuning process was robust and yielded a generalizable model, the BO process was configured to minimize the mean MAE derived from a 5-fold cross-validation ($K=5$) procedure, rather than using a single static validation subset.

The search space included the number of dense layers (ranging from 1 to 4), the number of units in each layer (from 16 to 128), the learning rate (from 0.0001 to 0.01), and the dropout rate (from 0.1 to 0.5). A total of 50 iterations were performed for the BO algorithm to explore this space and converge on the optimal set of hyperparameters presented in Table IV. Compared to the inefficiency of traditional grid search, BO intelligently guides the selection process by constructing a probabilistic surrogate model of the objective function. This approach not only ensures comprehensive exploration but also considerably reduces computation time. Its core process includes: (1) building a probabilistic surrogate model of the objective function based on historical evaluation results; (2) using the surrogate model to identify the best-performing combination of hyperparameters; (3) apply these selected hyperparameters to the true objective function, which involved training and validating the DNN 5 separate times (5-fold CV) to obtain a stable mean MAE score; (4) updating the surrogate model with the new result; and (5) iterating steps 2 through 4 until a predefined maximum number of iterations is reached. The resulting hyperparameter configuration effectively minimizes the network's loss function.

Furthermore, after determining the optimal hyperparameters, the final model's performance was rigorously evaluated using this 5-fold cross-validation ($K=5$) strategy. The detailed performance metrics (R^2 and MAE) for each of the 5 folds, along with the final averaged results, are provided in the Supplementary Material (Section S-2).

D. DDPG-based Optimization Framework

The parametric modeling process and the DNN-based surrogate model provide the foundation for the Integrated Optimization Framework linking urban morphology and energy performance. To achieve the MOO of these indicators, a DRL framework was developed, as shown in Fig. 2. This framework is composed of two parts. The first part provides a mathematical description of the problem by defining the Environment, a feasible Action Space, and the State Representation. The second part constructs a DDPG model that generates an Agent capable of stable and efficient learning. This Agent is closely coupled with the predefined Environment and utilizes an Experience Replay Buffer to iteratively optimize its Action Policy while evaluating overall performance. The core mechanism of this DRL framework is a closed-loop process: the Agent continuously collects feedback from the Environment, adaptively adjusts its strategy, and dynamically optimizes its actions to maximize a Reward Function designed to reflect the physical performance of the block. Ultimately, this agent-action-environment interaction solves the complex decision-making problem inherent in block design, enabling the practice of energy-efficient urban design and the creation of sustainable urban environments.

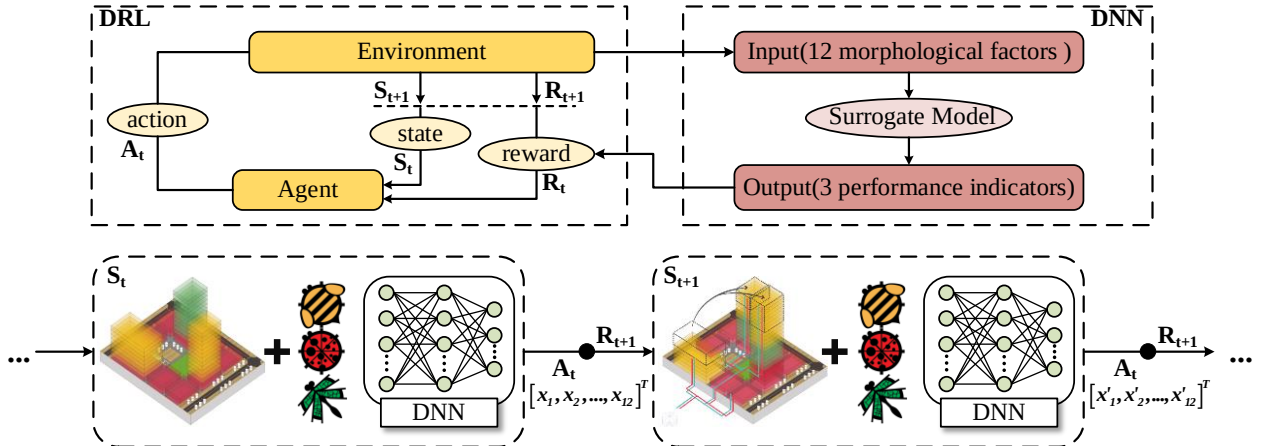


Fig. 2 Conceptual framework of the DRL-based optimization, illustrating the problem formulation as a Markov Decision Process

1) Problem Formulation

To address the MOO challenges in energy performance, this study constructs a DRL framework that is robust, adaptive, and computationally efficient. The core mechanism of this framework is a DRL agent that continuously receives State and Reward signals from its interaction with the environment. Based on this feedback, the agent iteratively updates its policy to autonomously learn and optimize urban morphology solutions that adapt to specific constraints and objectives. The agent, typically parameterized by a neural network, interacts with the environment through an exploratory trial-and-error process, learning an optimal behavioral policy based on the received rewards. In response, the virtual environment undergoes a State Transition according to the Action selected by the agent. This closed-loop dynamic feedback process accurately simulates the physical laws inherent in real-world engineering problems.

In this case study, the virtual environment consists of the block morphological factors and the DNN-based surrogate model. The DRL framework aims to explore and identify Pareto-optimal or near-optimal combinations of design strategies for various urban morphologies, providing technical support and a theoretical reference for energy-efficient design and sustainable urban development. Mathematically, this optimization process can be abstracted as a Markov Decision Process (MDP), which is typically defined by a four-tuple: (S, A, P, R) . In this tuple, S represents the State Space, A represents the Action Space, P is the State Transition Probability Function, and R is the Reward Function.

(1) *Action Space (A)*. In the DRL framework, an A is the fundamental unit of interaction between the Agent and the

Environment. By executing an action, denoted as $a \in A$, the Agent induces a change in the Environment's state. The core objective of the Agent is to learn an optimal policy, $\pi^*(s)$, which selects or generates an action a for each state s to maximize the expected cumulative reward. In this study, an action is defined as an adjustment to a set of preset building design parameters. Specifically, A consists of a combination of twelve key design parameters, each of which can be adjusted within a predefined range. Through its policy learning mechanism, the DRL framework can effectively explore the complex, high-dimensional action space to learn and generate sequences of actions that guide the design toward improved performance. As shown in Equation (2), a specific action a represents a particular configuration of these 12 parameters:

$$a = [x_1, x_2, \dots, x_{12}]^T \quad (2)$$

where a represents a specific action instance selected by the Agent, and x_i is the selected value for the i -th design parameter.

(2) *State Space(S)*. A State is an abstract representation of the Environment at a specific point in time, encapsulating the key information required for the Agent's decision-making process. Ideally, a state $s \in S$ should satisfy the Markov Property, meaning that the current state contains all historical information necessary to predict the future evolution of the system. The state serves as the input through which the Agent perceives the Environment and forms the basis for formulating subsequent action policies.

In this study, S is defined as the vector space composed of the key EPIs. Specifically, a state s directly corresponds to the three optimization objective values output by the DNN-based surrogate model. As shown in equation (3), the s is the result obtained from the surrogate model after a is executed:

$$s = [y_1, y_2, y_3]^T = [DNN_{EUI}(a), DNN_{EG}(a), DNN_H(a)] \quad (3)$$

where, s represents an instance of the environment's state, which is the performance vector resulting from action a ; y_1, y_2, y_3 are the predicted values for the three objectives, respectively; $DNN_{EUI}, DNN_{EG}, DNN_H$ is the output vector of EPIs from the surrogate model given action a as input.

(3) *State Transition Probability (P)*. The state transition probability, $P(s'|s, a)$, defines the probability distribution of the Environment transitioning to the next state s' given the current s and a . It describes the inherent dynamics of the Environment and its mechanism for responding to the Agent's actions. In the model-free DRL method employed in this study, P is typically unknown and does not require explicit modeling, as the Agent learns directly from samples obtained through interaction. However, the transition dynamics still form the basis of the learning process. In our framework, given an a , the DNN-based surrogate model deterministically or near-deterministically outputs the corresponding performance indicators, which in turn defines the next state.

(4) *Reward Function (R)*. The Reward Function, $R(s, a, s')$, is a core element of the DRL framework, providing a scalar feedback signal that guides the Agent's learning process. The Agent's objective is to maximize a long-term cumulative reward, typically expressed as the Expected Cumulative Discounted Reward. This study requires the simultaneous optimization of three performance objectives with different optimization directions: minimizing total energy use intensity (EUI, y_1), maximizing energy generation from PV (EG, y_2), and maximizing sunlight duration (H, y_3). The design of the reward function must integrate these conflicting objectives and guide the Agent toward exploring the Pareto Front. Therefore, we employ a reward function design based on a normalized weighted distance. This method first defines a Utopian Point that contains the ideal values for each objective. The problem is then transformed from minimizing the distance between the current state and this Utopian Point into a single-objective reinforcement learning problem focused on maximizing a scalar reward value.

1) *Normalization*: First, the three objectives are normalized by mapping them to a uniform $[0, 1]$ interval to eliminate the effects of differing scales.

$$\begin{aligned} y_{1,norm} &= (y_1 - y_{1min}) / (y_{1max} - y_{1min}) \\ y_{2,norm} &= (y_2 - y_{2min}) / (y_{2max} - y_{2min}) \\ y_{3,norm} &= (y_3 - y_{3min}) / (y_{3max} - y_{3min}) \end{aligned} \quad (4)$$

where y_{imin} and y_{imax} represent the minimum and maximum values of objective i within the feasible domain, respectively. In this normalized space, the utopian point is defined as $s_{u,norm} = [0, 1, 1]$, which represents the state where all objectives simultaneously reach their optimal values.

2) *Weighted distance calculation*: This step calculates the weighted Euclidean distance, $d_{weighted}$, from the normalized vector of the current state, $s_{norm} = [y_{1,norm}, y_{2,norm}, y_{3,norm}]$, to the normalized utopian point.

$$d_{weighted} = \sqrt{(w_{1,norm} - 0)^2 + (w_{2,norm} - 1)^2 + (w_{3,norm} - 1)^2} \quad (5)$$

where w_1, w_2, w_3 are preset, non-negative weight coefficients, such that $\sum w_i = 1$.

3) *Reward Function establishment*: As shown in Equation (6), the reward function uses the weighted Euclidean distance between the current state and the utopian point as the decision criterion. The reward function is designed with an upper limit of 10^6 to ensure convergence toward the Pareto-optimal front in practical applications.

$$R = 10^6 - d_{weighted} \quad (6)$$

where R represents the reward value.

To solve this formulated MDP, we selected the DDPG algorithm. DDPG is a model-free, actor-critic algorithm that is well-suited for this problem as it can effectively learn policies in high-dimensional, continuous action spaces. The algorithm works by concurrently training an "actor" network, which selects the optimal action (i.e., the morphological parameters), and a "critic" network, which evaluates the value of that action based on the reward function. The detailed background equations and standard pseudocode for the DDPG algorithm are provided in the Supplementary Material (Section S-3) for completeness. The specific implementation of our DDPG agents, which forms the core of our optimization engine, is detailed below.

2) DDPG Agent Implementation

The DDPG algorithm, which integrates a deterministic policy gradient with an actor-critic architecture, forms the core of our optimization engine. Its implementation is structured around a key methodological innovation for this study: a multi-scenario analysis designed to investigate the agent's ability to generate targeted design solutions based on varying decision-maker preferences. This is achieved by training multiple, independent DDPG agents, each guided by a unique reward function reflecting a specific design priority. This section details both the common architectural foundations of these agents and the specific configurations for each optimization scenario.

(a) Architectural and Foundational Parameters

The selection of appropriate hyperparameters is critical for the stability and performance of the DDPG algorithm. To ensure a fair and rigorous comparison between the three scenarios, a common set of foundational hyperparameters was established for all agents, with the only variable being the reward function weights that define their unique objectives. These carefully tuned parameters, summarized in Table III, are detailed below.

TABLE III
HYPERPARAMETERS SETTINGS FOR THE DDPG AGENT

Parameter	Value
Reward Discount Factor (γ)	0.999
Learning Rate (Actor)	0.001
Learning Rate (Critic)	0.002
Layers (Actor Network)	2
Units per Layer (Actor)	64, 32
Layers (Critic Network)	2
Units per Layer (Critic)	64, 32
Soft Update Factor (τ)	0.001
Batch Size	128
Replay Buffer Size	1e6
Maximum Episodes	200
Max Steps per Episode	40
Noise Decay Factor	0.9998

(1) Network Architecture and Learning Rates: As specified in Table III, both the Actor and Critic networks were constructed with two hidden layers of 64 and 32 neurons, respectively. This relatively simple architecture was found to be effective for this problem's complexity, avoiding the instability sometimes associated with deeper networks. The learning rates were set to 0.001 for the Actor and 0.002 for the Critic. A slightly higher learning rate for the Critic allows the value function to update faster, providing a more stable and accurate signal for the more slowly updating policy of the Actor.

(2) Reward Discount Factor (γ): The discount factor was set to 0.999. A value close to 1 encourages the agent to adopt a far-sighted strategy, prioritizing long-term cumulative rewards over immediate gains. This is highly suitable for a design optimization task, where the final performance of the generated morphology is the ultimate goal.

(3) Exploration Strategy: To ensure thorough exploration of the vast design space, an Ornstein-Uhlenbeck noise process was initially considered but a simpler Gaussian noise was found to be sufficient and more stable. Noise with an initial variance of 1.0 was added to the actor's actions. This variance was gradually reduced over time with a decay factor of 0.9998 at each step. This annealing strategy facilitates a smooth transition from a broad, exploratory search in the early stages of training to a more refined, exploitative search for the optimal policy as the agent gains more experience.

(4) Experience Replay and Batch Size: A large replay buffer with a capacity of 1 million transitions was utilized. This allows the agent to maintain a diverse and decorrelated set of past experiences, which is crucial for breaking temporal correlations and stabilizing the learning process. For each training step, a minibatch of 128 transitions was randomly sampled from this buffer to update the network weights.

By keeping these foundational parameters identical, we can confidently attribute any differences in the final optimized solutions directly to the influence of the scenario-specific reward function weights, which act as the sole differentiating factor in the agents' training directives.

(b) Multi-Scenario Implementation for Targeted Optimization

The central pillar of our implementation is the configuration of three distinct optimization scenarios, each designed to train a specialist agent. This is realized by adjusting the weight coefficients (w_1, w_2, w_3) within the reward function defined in Equation (5), a technique rooted in multi-criteria decision analysis (MCDA). By manipulating these weights, we can steer the agent's learning process toward different high-performance regions of the objective space. The three scenarios are defined as follows:

- Scenario 1: Balanced Performance. In this scenario, we assign equal weights ($w_1 = w_2 = w_3 = 1/3$). This configuration represents an unbiased or non-informative preference, tasking the agent with finding solutions that represent the best overall compromise among minimizing EUI_t , maximizing EG , and maximizing H . The agent trained under this scenario is expected to identify holistically superior designs.
- Scenario 2: Energy Saving Focus. Here, we prioritize the reduction of energy consumption by setting the weights to ($w_1 = 0.6$, $w_2 = w_3 = 0.2$). This scenario simulates a decision-maker whose primary concern is operational energy efficiency, such as in regions with high energy costs or stringent building energy codes. The agent is expected to converge on solutions that excel in minimizing EUI_t , potentially at the expense of the other two objectives.
- Scenario 3: Energy Generation Focus. To simulate a preference for maximizing on-site renewable energy production, we set the weights to ($w_1 = w_3 = 0.2, w_2 = 0.6$). This scenario is relevant for projects aiming for near-zero energy status or seeking to maximize revenue from electricity export. This agent is trained to discover morphologies with the highest potential for EG , even if it leads to a slight compromise in energy consumption or sunlight duration.

By training a separate agent for each of these scenarios, we transform the DRL framework from a single-solution optimizer into a flexible and powerful decision-support tool. It allows for an a priori articulation of design intent, which the agents then translate into targeted sets of high-quality design solutions. The overall architecture of the learning algorithm for each agent is illustrated in Fig. 3.

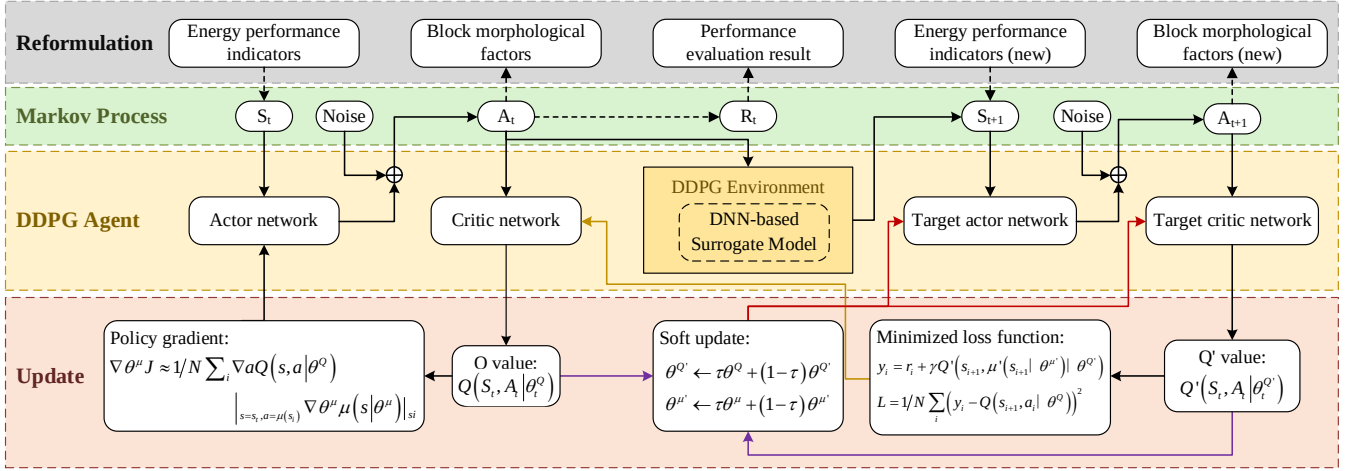


Fig. 3 Architecture of the DDPG agent, showing the Actor-Critic structure and learning process

III. RESULTS AND ANALYSIS

A. Results of surrogate model

Based on a dataset of simulations generated through the Grasshopper process, a DNN surrogate model was constructed to capture the complex relationships between the twelve design parameters and the three objective values. To ensure the robustness of the model's training and evaluation, the dataset was first randomly shuffled to eliminate sequential bias. It was then randomly partitioned into a training set and a testing set at a 4:1 ratio, with a fixed random seed to guarantee the reproducibility of the data split. Specifically, 400 data sets were used as the training subset for regression fitting and model parameter optimization, while the remaining 100 sets (20% of the total) were used as the testing subset to evaluate the model's predictive performance on unseen data. To avoid common issues such as overfitting and training instability, early stopping and dropout techniques were employed. As hyperparameters have a substantial impact on the model's predictive accuracy, BO was used for automated hyperparameter tuning. Table IV lists the optimal hyperparameter configuration for the DNN model as determined through the BO process.

TABLE IV
OPTIMAL HYPERPARAMETERS OF DNN DETERMINED BY BO

Hyper parameter	Value
Number of dense layers	2
Units of 1st layer	64
Units of 2nd layer	32
Dropout rate	0.2
Learning rate	0.001
Batch Size	32
Optimizer	Adam
Activation	ReLU

To ensure a rigorous evaluation of the model's generalization capability and avoid potential biases from a single train/test split, a 5-fold cross-validation ($K=5$) strategy was employed. The dataset was divided into 5 equal folds (100 samples each). The model was trained 5 times; in each iteration, 4 folds (400 samples) were used for training, and the held-out 1-fold (100

samples) was used for testing.

The coefficient of determination (R^2) and Mean Absolute Error (MAE) were used as the key evaluation metrics. Table V summarizes the mean and standard deviation of these metrics across the 5-fold cross-validation. Specifically, R^2 measures the model's goodness-of-fit and reflects the extent to which it explains the variance in the data. The MAE represents the absolute prediction error and is independent of the magnitude of the objective values, which facilitates comparisons across different objectives. The evaluation results show that the R^2 values for all objectives in both the training and testing sets are close to 1, indicating an excellent model fit.

Concurrently, all average MAE values are low, reflecting small prediction errors. More importantly, the minimal standard deviation across the folds indicates that the model's performance is highly robust and not dependent on a specific data partition. This confirms the model's reliability and that it does not suffer from overfitting. Taken together, these results lead to the conclusion that the constructed DNN surrogate model provides accurate and reliable predictions of building performance, offering robust support for the associated design optimization. The detailed results for each of the 5 folds are provided in the Supplementary Material.

TABLE V
SUMMARY OF DNN PREDICTION PERFORMANCE USING 5-FOLD CROSS-VALIDATION

Objective	Mean Value (K=5)		Standard Deviation	
	R^2	MAE	R^2	MAE
EUI_t	0.973	0.031	± 0.011	± 0.007
EG	0.969	0.04	± 0.014	± 0.005
H	0.971	0.028	± 0.009	± 0.006

To visually demonstrate the predictive performance, Fig. 4 and Fig. 5 present the prediction results of the DNN surrogate model and its error distribution, respectively, for a representative test fold. In Fig. 4, the green curves representing the predicted values and the gray curves representing the simulated values almost completely overlap. This indicates that the DNN model can synchronously predict the three objective values with high precision across multiple combinations of design parameters, demonstrating an excellent goodness-of-fit. Fig. 5 provides a statistical description of the prediction errors using a normal distribution curve. The results show that nearly all error values fall within the 2σ range, which confirms the model's excellent error-control capabilities during the training phase. These characteristics of high precision and low error ensure the predictive reliability of the DNN surrogate model, providing a robust foundation for the subsequent DRL-based multi-objective optimization of block performance.

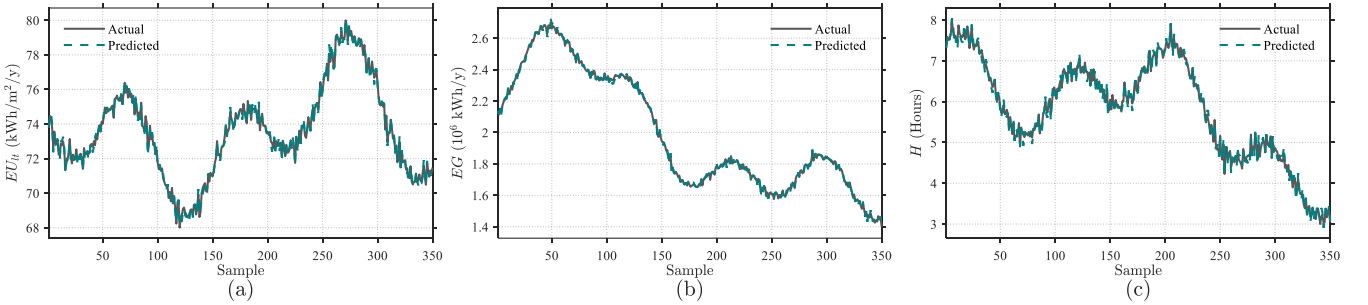


Fig. 4. DNN-based prediction results for the three objectives on the typical fold: (a) EUI_t (y_1); (b) EG (y_2); (c) H (y_3)

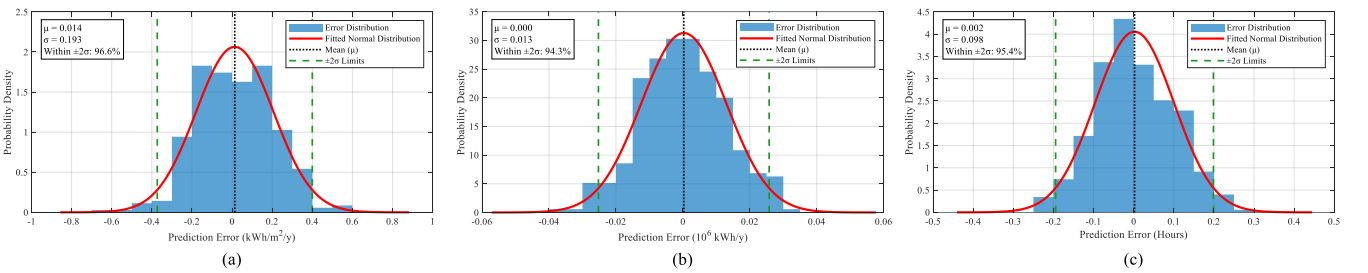


Fig. 5. Error distribution of DNN-based prediction results for the three objectives on the typical fold: (a) EUI_t (y_1); (b) EG (y_2); (c) H (y_3)

B. DDPG learning curve

The learning dynamics of the agents were monitored to ensure convergence to a stable and effective policy. To illustrate this process, this study presents a typical, representative training run from a DDPG agent operating under Scenario 1. As shown in Fig. 6, this run was selected for demonstration as the agents from the other two scenarios exhibited highly similar convergence patterns, which validates the overall robustness of the DRL framework. The learning process, as depicted in the four subplots of Fig. 6, can be analyzed in detail: Cumulative Reward (Fig. 6a): The primary indicator of learning, the cumulative reward, shows a steep and consistent increase during the first 500-1000 iterations. This signifies that the agent is rapidly learning an effective policy to maximize its reward by finding morphologies closer to the utopian point. After this initial phase, the reward curve enters a plateau, indicating that the policy has converged to a stable, high-performance state. Objective Value Trajectories (Fig. 6b, Fig. 6c, Fig. 6d): The trajectories of the individual objectives provide deeper insight

into the agent's decision-making process. In the early exploration phase, the values for EUI_t (y_1), EG (y_2), and H (y_3) show wide fluctuations as the agent experiments with diverse and often suboptimal morphological configurations. As training progresses and the policy improves, these fluctuations narrow significantly, and the values converge to a stable mean. For the Balanced agent, the final converged values represent a strong compromise across all three metrics, hovering around 71.6 kWh/m²/y for EUI_t , 2.40 x 10⁶ kWh/y for EG , and 7.27 h for H .

While demonstrating similar convergence patterns, the agents for the other two scenarios stabilized at different objective value ranges, consistent with their specialized reward functions. For instance, the Scenario 2 agent's learning curve for EUI_t converged to a markedly lower average value of approximately 69.8 kWh/m²/y. Conversely, the Scenario 3 agent's trajectory for EG stabilized at a significantly higher level of approximately 2.65 x 10⁶ kWh/y. The successful and distinct convergence of all three agents provides strong evidence that the DRL framework can reliably train specialist agents capable of mastering different, preference-driven optimization tasks.

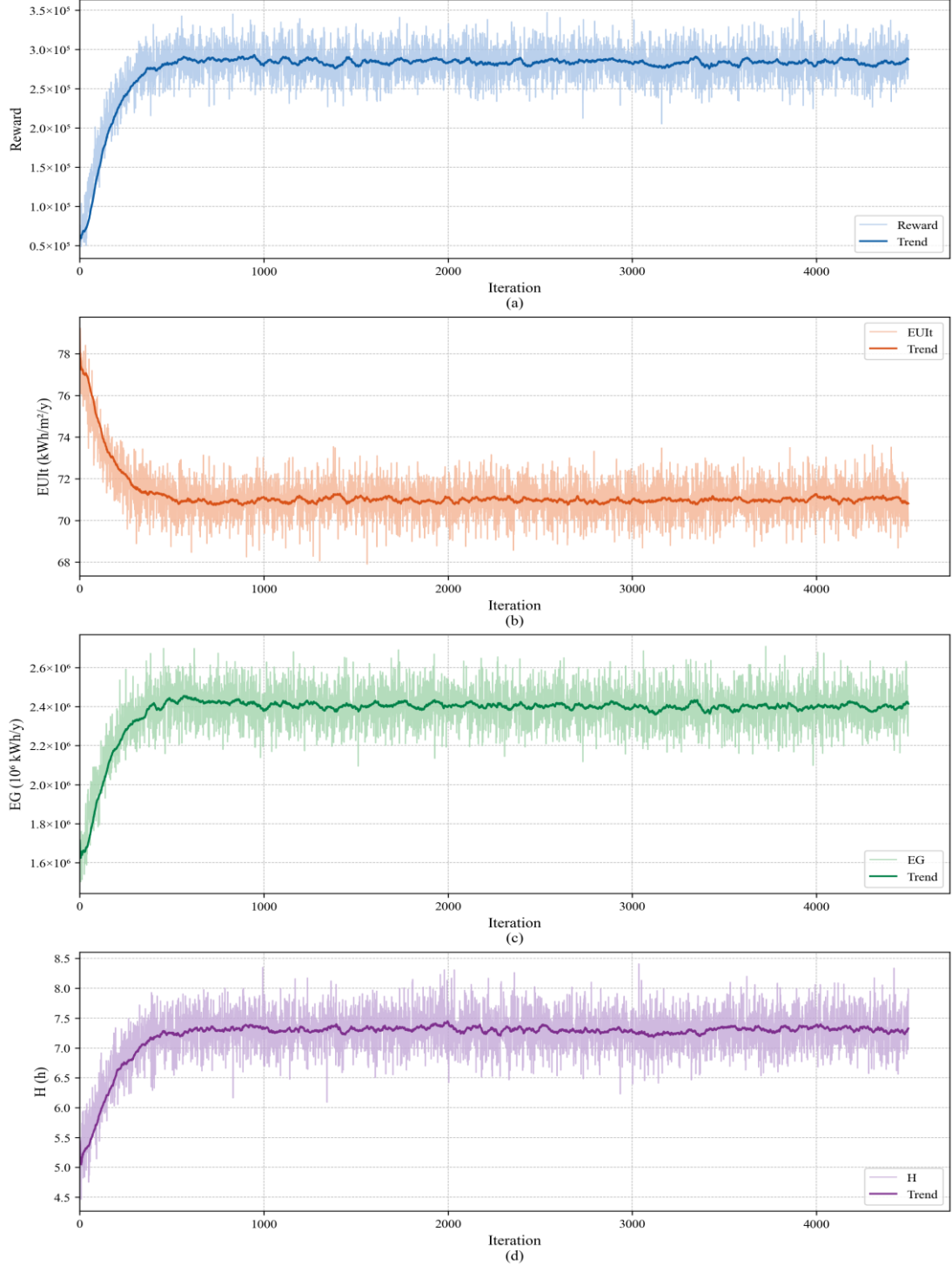


Fig. 6. DDPG learning curve during the training process

C. Results of the multi-objective optimization

Before presenting the comparative results, it is essential to clarify the methodology used to generate the DRL solution

clusters. To ensure the robustness and statistical validity of the findings, each of the three DRL scenarios was trained independently and repeatedly, each time with a different random seed. This process produces a distribution of high-performance solutions for each agent, reflecting the stable convergence region of its learned policy. This rigorous approach allows for a statistical comparison of not just the average performance, but also the stability and reliability of the solutions generated by each method. This analysis will first explore the distinct performance and strategies of these three DRL scenarios, before comparing them comprehensively against the NSGA-II benchmark.

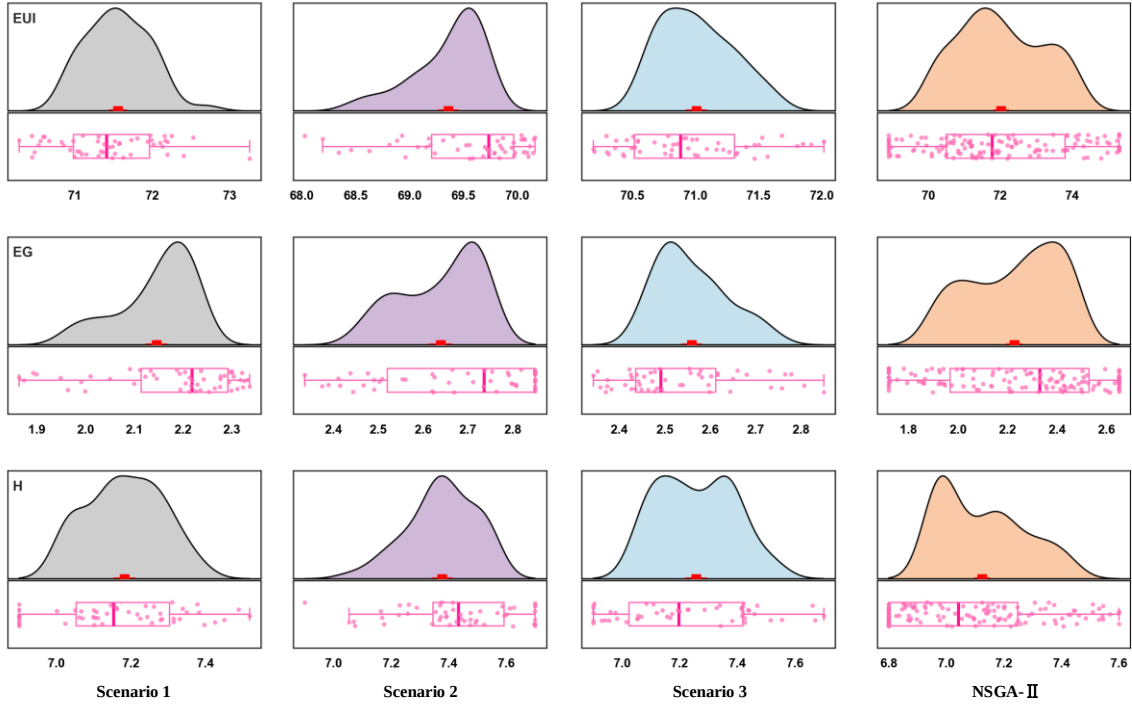


Fig. 7. Statistical comparison of solution distributions for the three objectives across the four optimization methods

1) Analysis of DRL Multi-Scenario Performance

The primary strength of the DRL framework resides in its flexibility to generate targeted design solutions. By specifying a design preference a priori through the reward function weights, the framework effectively trains three distinct "specialist agents," each possessing a unique learned policy. The statistical outcomes of these policies appear in Fig. 7, which illustrates the nuanced differences among the scenarios. As intended, Scenario 2 achieves the lowest median EUI with a sharp, uni-modal distribution, confirming its effectiveness in identifying stable, low-energy solutions. Scenario 3 successfully shifts its distribution to attain a high median EG and H, surpassing Scenario 1 in these metrics. Scenario 1 arrives at a compromise solution. Notably, the most compelling and non-obvious finding is that the Scenario 2 agent identifies a dominant sweet-spot region. It not only achieves the lowest median EUI but also unexpectedly secures the highest median EG and the highest median H, outperforming all other methods across the three objectives.

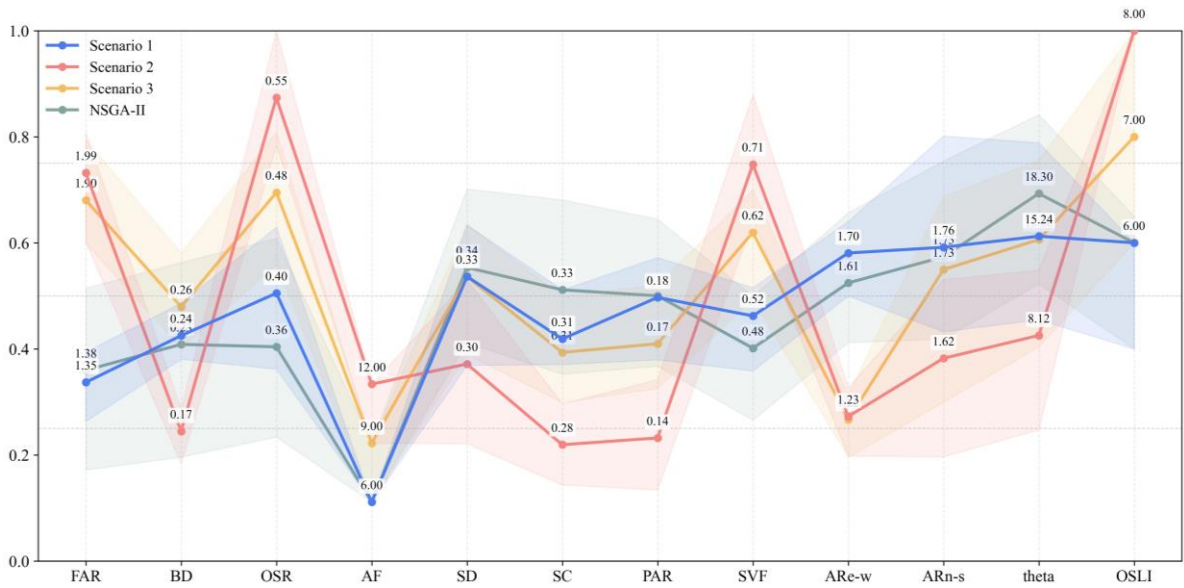


Fig. 8. Parallel coordinate plot comparing the learned design strategies for the 12 UMFs across the four optimization methods

These divergent performance outcomes are a direct result of the agents learning distinct design strategies, which are captured in Fig. 8. This plot clearly illustrates the different design fingerprints of the agents. A clear divergence in strategy is visible:

the dominant Scenario 2 learned to favor a unique combination of significantly lower *BD* and a dramatically higher *OSR* and *SVF*. In contrast, Scenario 3 learned to prioritize a higher *FAR* and a different *OSLI* to maximize generation potential. Scenario 1 learned a more average policy, balancing these factors without specializing. This demonstrates that the DRL agents are not just finding random solutions, but are learning and converging upon specific, intelligible, and different morphological strategies based on their given preference. This strategic specialization is also visualized in Fig. 11. Each DRL agent's flower exhibits a unique morphological petal structure. For example, the Scenario 2 flower is dominated by an exceptionally long petal for *OSR*, while the Scenario 3 flower shows a more extended *FAR* petal.

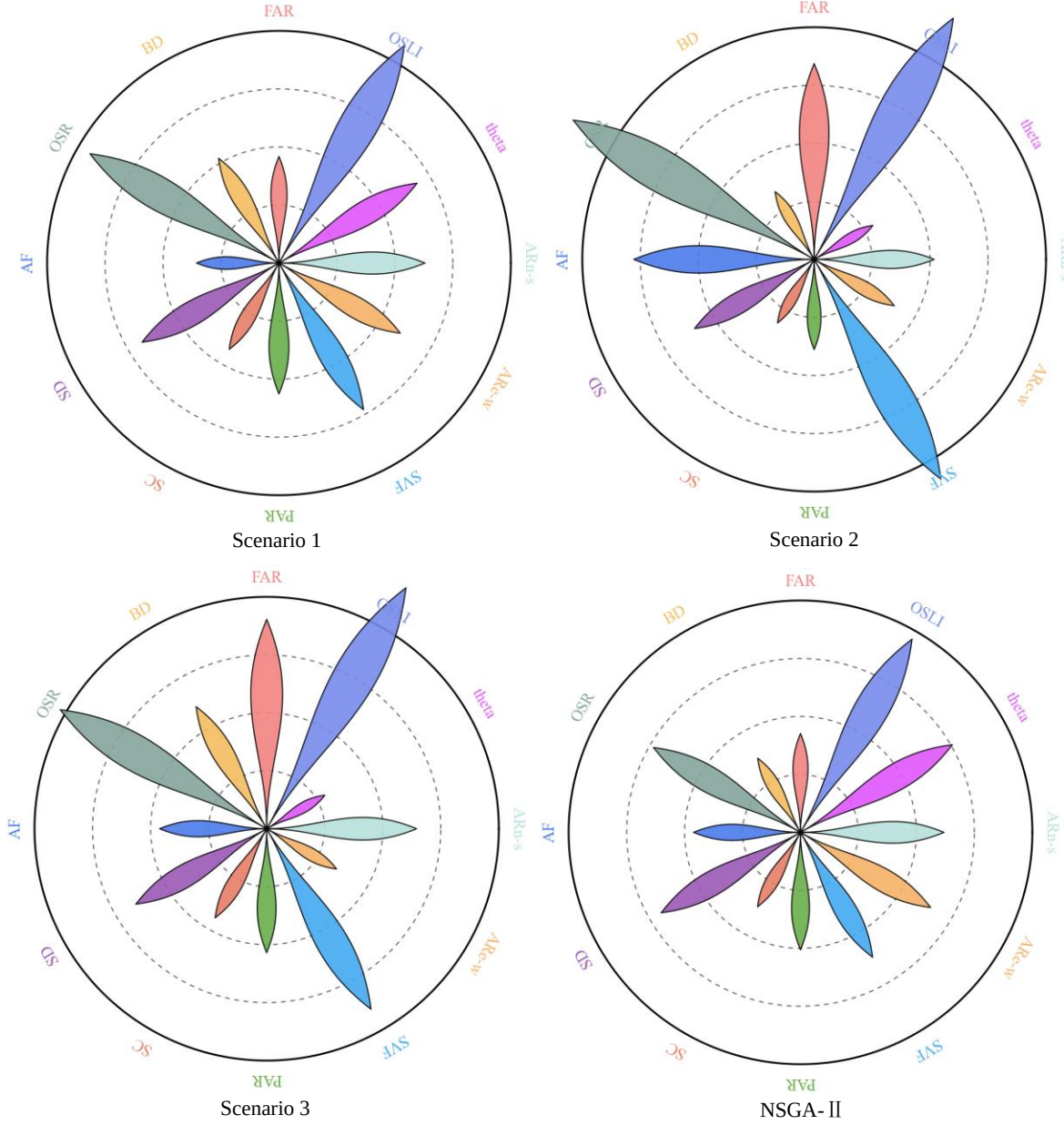


Fig. 9. Visualization of learned morphological strategies for each optimization method

2) Comparative Analysis against NSGA-II

After establishing the specialization and internal performance of the DRL agents, we compare them to the NSGA-II benchmark. Fig. 10 clearly shows the quantitative superiority of the DRL solution sets. In all three DRL scenarios, both Hypervolume and Inverted Generational Distance metrics surpass the benchmark. Scenario 2 stands out as the leader, achieving the highest mean Hypervolume and the lowest mean Inverted Generational Distance. This indicates that its solution set is closer to the true Pareto front and occupies a larger volume of the high-performance space. In contrast, the NSGA-II solution set results in the lowest mean Hypervolume and the highest mean Inverted Generational Distance. The narrow error bars for the DRL scenarios, particularly for Scenario 2, underscore their stability and reliability, unlike the high variance observed in the NSGA-II results.

Fig. 7 illustrates the quantitative dominance of DRL agents over the NSGA-II solution set by comparing their performance across three objectives. DRL agents surpass NSGA-II in all aspects: they achieve the lowest median EUI, the highest median EG, and the highest median H. The NSGA-II's performance is hampered by its wide, bimodal EUI distribution, indicating a failure to converge and a split between two suboptimal strategies. This high variance and lack of convergence render NSGA-

II's solutions unreliable and less effective for designers. Fig. 11 visually summarizes this comprehensive dominance by plotting all three objectives on a normalized scale, allowing for an immediate comparison. Scenario 2's superiority is clear: its boxplots show the lowest EUIt and the highest EG and H. In contrast, NSGA-II's boxplots depict the worst median performance across all objectives and the longest whiskers, confirming the extreme variance and unreliability highlighted in the density plots.

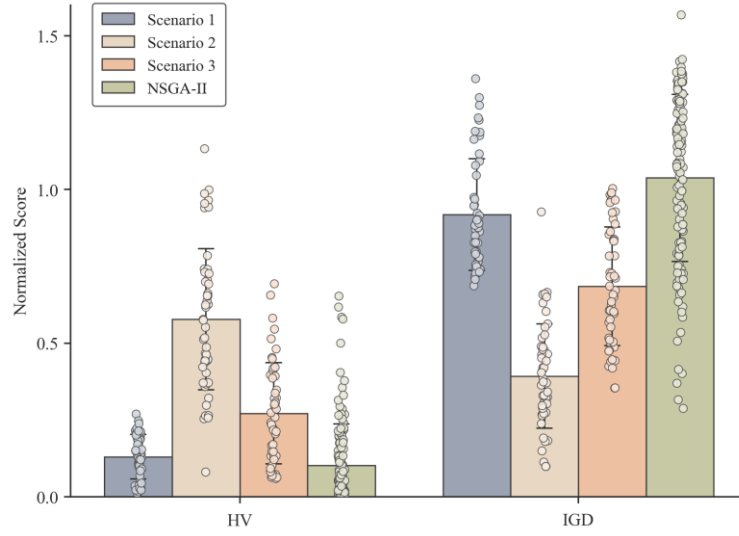


Fig. 10. Quantitative comparison of solution set quality using Hypervolume and Inverted Generational Distance metrics

The performance gap stems from the strategies employed. Upon revisiting Fig. 8, the NSGA-II strategy seems indecisive. Its median values typically fall between those of the DRL specialists, lacking a distinct, optimized fingerprint. Notably, its interquartile ranges for crucial variables such as *OSR*, *SVF*, and θ are extensive. This observation supports previous plots, indicating that NSGA-II is unfocused, yielding a broad, varied, and ultimately ineffective set of designs. Designers receive no clear strategic guidance. Fig. 9 visually reinforces this, as the NSGA-II flower displays a more balanced but less specialized shape, missing the dominant, strategic spikes that the DRL agents developed. In summary, the DRL framework not only discovered better solutions but also learned superior, specialized strategies that the generalized NSGA-II search did not identify.

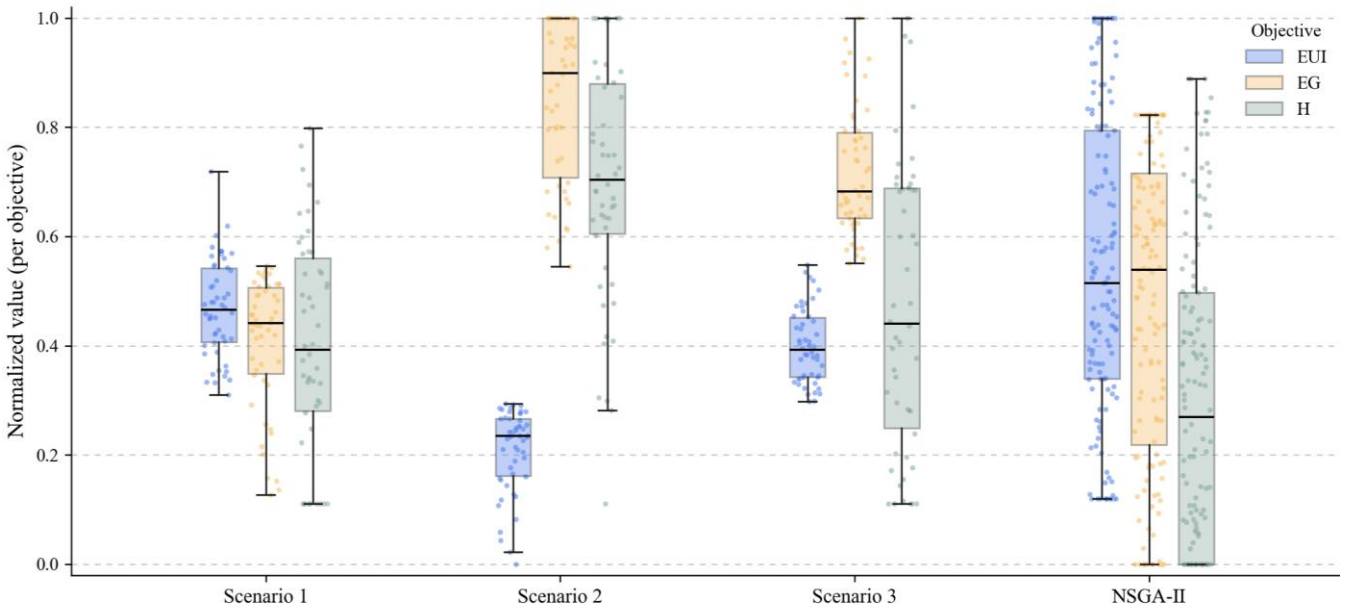


Fig. 11. Normalized boxplot comparison of optimization objectives

3) Correlation analysis

To quantitatively grasp the linear relationships between the 12 UMFs and the three performance objectives, we conducted a Pearson correlation analysis. The findings are presented in two formats: a comprehensive heatmap for a high-level overview and a series of joint distribution plots for a detailed statistical examination of the most crucial variable pairs. The overall correlation matrix appears in Fig. 12. This heatmap illustrates the direction and relative strength of each factor's influence on *EUIt*, *EG*, and *H*. Additionally, the bubble plot at the bottom aggregates the average absolute correlation, highlighting which factors, such as *FAR*, *OSR*, and *PAR*, exert the most significant overall impact across all objectives.

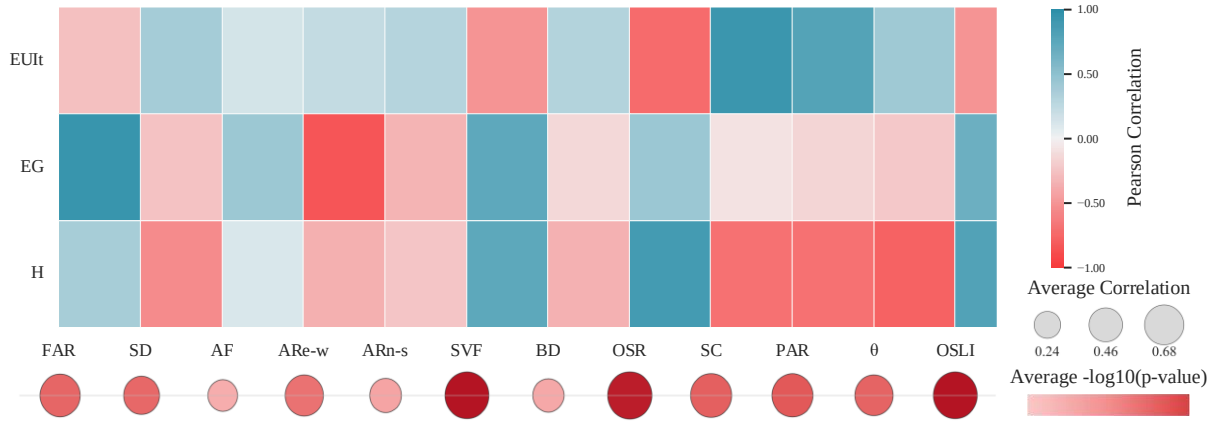


Fig. 12. Heatmap of Correlation Between UMFs and Performance Targets

Fig. 13 provides a comprehensive joint distribution analysis to explore significant relationships and their distributions. This analysis highlights several key performance drivers. For *EUIt*, two factors display strong, contrasting correlations. *PAR* has a strong positive correlation (0.796), indicating that less compact and more articulated building shapes (high *PAR*) expose more facade area, leading to increased thermal losses or gains, which in turn raises energy use. In contrast, *OSR* shows a strong negative correlation (-0.722), suggesting that increased open space enhances natural ventilation and daylight access, thus reducing the need for mechanical cooling and lighting. For *EG*, the analysis identifies *FAR* as the strongest driver, with a robust positive correlation (0.935). This suggests that denser developments, which accommodate a larger total rooftop area, significantly boost the potential for on-site PV generation. Additionally, *SVF* demonstrates a strong positive correlation (0.740), indicating that less obstruction from neighboring buildings enhances solar access for PV panels. For *H*: The relationships are clear and intuitive. *OSLI* (0.804) and *SVF* (0.746) emerge as the strongest positive factors, underscoring that sky access and minimal obstructions are crucial for maximizing pedestrian-level daylight. Conversely, Block Orientation (-0.770) and *PAR* (-0.693) exhibit strong negative correlations, indicating that orientations far from due south and complex building perimeters, which cast long shadows, hinder sunlight access.

These findings validate the physical principles underlying the simulation and offer a clear rationale for the optimal design strategies identified by the DRL agents in the previous section. For example, the Scenario 2 agent's sweet-spot strategy, characterized by low *BD*, high *OSR*, and high *SVF*, is strongly supported by this correlation analysis. These factors simultaneously decrease *EUIt* while increasing *EG* and *H*.

IV. DISCUSSION

This study introduces a novel DRL-DNN integrated framework to tackle the intricate co-optimization of urban morphology and energy performance. The findings not only validate the framework's effectiveness but also offer insights that challenge traditional views of this MOO problem. This discussion synthesizes the main findings, contextualizes their implications, and addresses the study's limitations.

TABLE VI
PERFORMANCE METRIC COMPARISON OF OPTIMAL SOLUTION CLUSTERS

Method	HV	IGD	EUIt	EG (10 ⁶ kWh/y)	H (h)
NSGA-II	0.1	1.04	72.5	2.2	7.1
Scenario 1	0.14	0.92	71.8	2.15	7.2
Scenario 3	0.25	0.68	71	2.5	7.3
Scenario 2	0.58	0.39	69.5	2.65	7.45

The most notable finding is the identification of a dominant solution region by the Scenario 2 agent. Traditional MOO methods, such as the NSGA-II benchmark, define a Pareto front, implying a set of trade-offs. Our NSGA-II results align with this, yielding a scattered, high-variance solution set with average performance across all objectives. However, the DRL agent trained in Scenario 2 did not merely identify a better trade-off; it pinpointed an optimal region in the high-dimensional design space. This morphological strategy simultaneously minimizes *EUIt*, maximizes *EG*, and maximizes *H*. This indicates that the objective space contains complex, non-linear valleys of high performance that traditional evolutionary algorithms, with their generalized search patterns, fail to detect. The DRL agent, by learning a policy through repeated interaction, effectively navigates this complex landscape and exploits a subtle design strategy that achieves comprehensive, multi-objective dominance. Quantitative evidence is presented in Table VI and Fig. 10, where Scenario 2's solution set is mathematically superior in both convergence and volume.

This study's multi-scenario design underscores the flexibility of the DRL framework as a decision-support tool. Unlike NSGA-II, which generates a single, undifferentiated Pareto front requiring post-processing by designers to identify desired solutions, our framework facilitates a priori preference articulation. The results demonstrate the effectiveness of this preference-setting. Agents in Scenario 2 and Scenario 3 develop distinct design fingerprints, as illustrated in Fig. 8, tailored to

their specific reward signals. For example, Scenario 3 accurately identifies a higher FAR as crucial for its generation-focused objective, a strategy that contrasts with Scenario 2's approach. This transformation shifts the optimization process from a blind search to an intelligent, goal-directed exploration, offering a practical solution for designers who need to create bespoke solutions aligned with specific project priorities. The DRL framework offers interpretable design insights, addressing the common black box criticism of optimization methods. Unlike the NSGA-II results, which present a diverse solution set with wide interquartile ranges across most design variables, the DRL agents converge on precise strategies, offering clear strategic guidance. The correlation analysis elucidates the success of the Scenario 2 agent. This agent adopts a strategy that prefers low *BD*, high *OSR*, and high *SVF*. The correlation findings support this approach, showing that this combination reduces *EUIt* while increasing *EG* and *H*. Thus, the DRL framework delivers not only solutions but also transferable design knowledge.

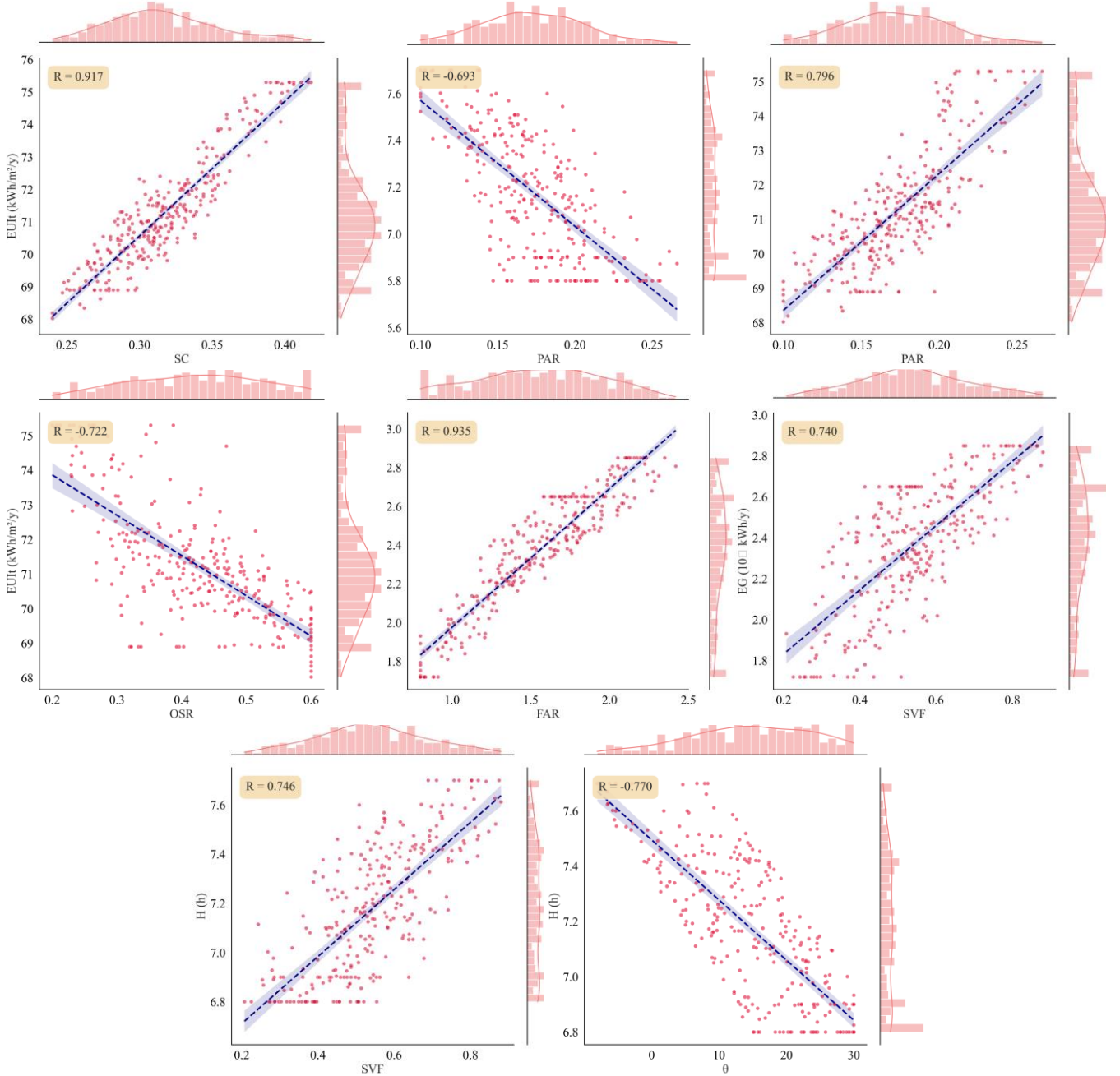


Fig. 13. Joint Distribution Plot of Key Morphological Factors and Performance Targets

The proposed framework offers significant advantages, yet several limitations require acknowledgment. First, the findings stem from a single case study within a specific climate zone. The "low-density, high-openness" strategy may not be applicable in cold climates, where compactness is preferred. Future research should validate the framework across various climate contexts to ensure its broader applicability. Second, although the surrogate model demonstrates high accuracy, it remains an approximation. Direct integration of the DRL agent with the simulation engine could reveal more nuanced insights, though this would entail a significantly higher computational cost. Lastly, the study employs a reward function with fixed weights. A more advanced approach could incorporate dynamic weighting or interactive human-in-the-loop feedback, enabling designers to adjust preferences during the training process. This enhancement could further increase the framework's utility as an interactive design tool.

V. CONCLUSION

This study introduces and validates an innovative framework utilizing DRL and DNN to tackle the intricate task of co-optimizing urban morphology and energy performance. Traditional methods, such as NSGA-II, often face challenges due to their computational intensity and inefficiency in navigating the high-dimensional, non-linear design space, resulting in suboptimal trade-offs. Our results reveal that the DRL framework effectively addresses these limitations, providing a more intelligent and adaptable approach to design optimization. The main conclusions are threefold: First, the DRL framework serves as a highly adaptable decision-support tool. By defining design priorities in advance using various reward functions, it effectively trains specialist agents. These agents adeptly explore the design space and converge on distinct, high-performance solution clusters, each tailored to specific objectives. Second, the DRL framework clearly outperforms the NSGA-II benchmark, and this advantage is both significant and comprehensive. In terms of metrics, DRL solution sets, especially from Scenario 2, exhibit superior HV and IGD values, indicating better convergence and diversity. Notably, the Scenario 2 agent identifies a dominant solution region that NSGA-II completely overlooks. This solution cluster excels across all three objectives: achieving the lowest *EUI*, the highest *EG*, and the highest *H*. This discovery challenges the traditional trade-off paradigm by suggesting that DRL can uncover non-obvious, dominant solutions that generalized algorithms fail to detect. Third, the DRL agents acquire interpretable and transferable design strategies. Unlike the black box nature of NSGA-II's scattered results, the dominant Scenario 2 agent develops a clear design fingerprint, favoring low building density, a high open-space ratio, and a high sky-view factor. Correlation analysis subsequently validates the success of this specific strategy. Consequently, the framework offers not only a set of solutions but also actionable design knowledge.

In summary, this paper transforms the optimization problem from a static blind search into an intelligent, goal-directed learning process. The DRL-DNN framework functions as a powerful, flexible tool that generates bespoke, high-performance, and interpretable design solutions. Future work should validate this framework across different climate zones and develop more advanced interactive reward functions. This study demonstrates that DRL has significant potential to move beyond simple optimization and to fundamentally enhance the workflow of sustainable urban design.

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